VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. IV YEAR
MECHANICAL ENGINEERING

VII SEMESTER **R22**

Course Code	Title of the Course	L	T	P/D	CH	C
22PC1ME401	Industry 5.0	3	0	0	3	3
22PC1ME402	Turbomachinery	3	0	0	3	3
	PROFESSIONAL ELECTIVE – III					
22PE1ME401	Computational Fluid Dynamics	3	0	0	3	3
22PE1ME402	Production Planning and Control					
22PE1ME403	Machine Learning and Its Applications in Mechanical Engineering					
22PE1ME404	Mechanics of Composite Materials					
22PE1ME405	Modern Machining Processes					
	PROFESSIONAL ELECTIVE – IV THROUGH SWAYAM-NPTEL					
	Design Stream	3	0	0	3	3
	Industrial Engineering and Management Stream					
	Manufacturing and Production Stream					
	Robotics Stream					
	Emerging Technologies Stream					
	OPEN ELECTIVE – III	3	0	0	3	3
22PC2ME411	Production Drawing Practices Laboratory	0	0	2	2	1
22PC2ME412	Computation and Simulation Laboratory	0	0	2	2	1
22PW4ME401	Major Project Phase – I	0	0	6	6	3
Total		**14**	**0**	**12**	**26**	**20**

L – Lecture T – Tutorial P – Practical D – Drawing CH – Contact Hours/Week
C – Credits SE – Sessional Examination CA – Class Assessment ELA – Experiential Learning Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation LR – Lab Record
CP – Course Project PE – Practical Examination

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. IV YEAR
MECHANICAL ENGINEERING

VIII SEMESTER **R22**

Course Code	Title of the Course	L	T	P/D	CH	C
	PROFESSIONAL ELECTIVE – V					
22PE1ME406	Gas Dynamics and Jet Propulsion	3	0	0	3	3
22PE1ME407	Design for Manufacturing and Assembly					
22PE1ME408	Advances in CAD/CAM					
22PE1ME409	Product Design and Process Planning					
22PE1ME410	Metal Casting Technology					
	PROFESSIONAL ELECTIVE – VI					
22PE1ME411	Power Plant Engineering	3	0	0	3	3
22PE1ME412	Sustainable Development Technologies					
22PE1ME413	Advanced Mechanics of Solids					
22PE1ME414	Electric and Autonomous Vehicles					
22PE1ME415	Computer Integrated Manufacturing					
	OPEN ELECTIVE – IV	3	0	0	3	3
22PW4ME402	Major Project Phase – II	0	0	22	22	11
Total		**9**	**0**	**22**	**31**	**20**

L – Lecture T – Tutorial P – Practical D – Drawing CH – Contact Hours/Week
C – Credits SE – Sessional Examination CA – Class Assessment ELA – Experiential Learning Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation LR – Lab Record
CP – Course Project PE – Practical Examination

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC1ME401) INDUSTRY 5.0

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Industrial Engineering and Management

COURSE OBJECTIVES:
- To understand the evolution of Industry 5.0
- To explore key technologies in Industry 5.0
- To explore the smart manufacturing strategies and the role of digital twins in smart industry
- To investigate the use of sustainability metrics and ESG frameworks in Industry 5.0
- To discuss the strategic challenges of Industry 5.0

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand the technological advancements that led from Industry 4.0 to Industry 5.0
CO-2: Implement and Integrate IIoT and Smart Technologies
CO-3: Apply the advanced Technologies in the context of Industry 5.0 for manufacturing
CO-4: Understand and address the ethical, policy, and strategic challenges of Industry 5.0

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	3	2	1	1	2	1	1	-	-	-	-	2	3	2
CO-2	3	3	3	2	3	2	3	-	-	-	1	2	2	3
CO-3	3	3	2	-	3	1	2	2	-	-	2	3	3	3
CO-4	1	2	1	3	2	2	1	2	1	1	3	2	1	2

UNIT-I:
Evolution and Foundations of Industry 5.0:
Historical Overview: Industrial Revolutions (1.0 to 5.0), Transition from Automation (Industry 4.0) to Collaboration (Industry 5.0), Key Drivers, Enablers, and Barriers of Industry 5.0 Adoption, Sustainability Metrics and Environmental, Social, and Governance (ESG) Frameworks, Comparative Analysis: Smart Factories of Industry 5.0 vs Traditional Factories, Leveraging Industrial Big Data and Predictive Analytics for Smart Business Operations

UNIT-II:
Industrial Internet of Things (IIOT) and Connected Enterprises: Core Concepts of Internet of Things (IoT) and IIoT, Role of Internet of Services (IoS) in Smart Manufacturing, Integration of IIoT for Operational Efficiency and Predictive

Maintenance, IIoT Architecture: Edge Computing, Cloud Integration, and Data Pipelines, Smart Factories: Real-time Monitoring, Smart Logistics, and Connected Devices, Enterprise Digital Twins: Simulating Industrial Processes for Optimization

UNIT-III:

Advanced Technologies Powering Industry 5.0: Cyber-Physical Systems (CPS) and Industrial Automation, Collaborative Robotics (Cobots) and Human-Machine Interaction (HMI), Artificial Intelligence (AI) and Machine Learning (ML) for Predictive Manufacturing, Industrial Cybersecurity: Threat Detection, Risk Management, and Data Integrity, Mobile and Edge Computing: Enabling Real-time Decision Making, Ethical AI and Responsible Technology Deployment, Agile and Distributed Supply Chains with AI-Driven Insights

UNIT-IV:

Smart Manufacturing and Industry 5.0 Implementation:

Green Manufacturing: Circular Economy, Resource Optimization, and Market Demands, Society 5.0 and Leadership 5.0: Human-Centric Industrial Strategies, AI-Enabled Production Systems and Smart Automation Tools, Intelligent Product Design and Development Using AI and IIoT, Real-World Applications and Case Studies.

UNIT-V:

Future Applications, Challenges, and Strategic Impact: Role of Ergonomics and Human-Centric Design in Industry 5.0, Future Trends in Smart Manufacturing and Adaptive Production Systems, Business Transformation: Digitalization and Sustainable Business Models, Strategic Challenges: Workforce Upskilling, Ethical Considerations, and Policy Frameworks, Emerging Industry Applications and Case Studies.

TEXT BOOKS:

1. Introduction to Industrial Internet of Things and Industry 4.0, Sudip Misra, Chandana Roy, Anandarup Mukherjee, CRC Press, 2020
2. Industry 5.0, The Future of the Industrial Economy, Uthayan Elangovan, 1st Edition, Taylor& Francis

REFERENCES:

1. Sustainability and Innovation in Manufacturing Enterprises: Indicators, Models and Assessment for Industry 5.0, Anca Draghici, Larisa Ivascu, Springer Nature, September 2021
2. Industry 4.0: The Industrial Internet of Things, Alasdair Gilchrist, A Press, 2016
3. Electronics in Advanced Research Industries: Industry 4.0 to Industry 5.0 Advances, Alessandro Massaro, Wiley-IEEE Press, 2021
4. Additive Manufacturing Technologies Rapid Prototyping to Direct Digital Manufacturing, Lan Gibson, David W. Rosen and Brent Stucker, Springer, 2010

ONLINE RESOURCES:

1. https://nptel.ac.in/courses/106105195
2. https://archive.nptel.ac.in/courses/110/106/110106146/

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC1ME402) TURBO MACHINERY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Basics of Thermodynamics, Basic Concepts of Thermal Engineering

COURSE OBJECTIVES:
- To analyze and understand various energy conversations that take place in a turbo machines
- To apply the principles of turbo machines
- To evaluate governing mathematical equations to perform theoretical calculations
- To create a model for condensers, compressors and turbines

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Analyze the working, performance of boilers and its accessories
CO-2: understand the performance of steam condensers and nozzles
CO-3: Evaluate the efficiency of steam turbines& Compressors
CO-4: Analyze the performance of gas turbines and working of jet propulsions

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	3	3	3	3	2	1	2	-	-	2	1	3	3	-
CO-2	3	3	3	3	2	1	2	-	-	2	1	3	3	-
CO-3	3	3	3	3	2	1	2	-	-	2	1	3	3	-
CO-4	3	3	3	3	2	1	2	-	-	2	1	3	3	-

UNIT-I:
Basic Concepts, Steam Generators:
Introduction, Classification of Boilers, Working Principles of Fire Tube Boilers -Cochran & Lancashire Boilers, Water Tube Boilers-Babcock and Wilcox & Stirling Boiler, High Pressure Boilers – Lamont & Benson Boilers, Boiler draught and performance of boilers, Equivalent evaporation.
Steam Condensers: Introduction, Classification of condensers – working principle of different types of condensers, vacuum efficiency and condenser efficiency – air leakage, sources and its affects, air pump- cooling water requirement.

UNIT-II:
Steam Nozzles: Functions of nozzle, applications, types, flow through nozzles, Thermodynamic analysis, assumptions, velocity at nozzle exit, Ideal and actual expansion in nozzle, velocity coefficient, condition for maximum discharge, nozzle

efficiency, Critical pressure ratio, Supersaturated flow and its effects, degree of super saturation, degree of under cooling, Wilson line.

UNIT-III:
Steam Turbines: Impulse Turbine: Mechanical details, velocity diagram, effect of friction, power developed, axial thrust, diagram efficiency, Condition for maximum efficiency, Methods to reduce rotor speed - velocity compounding, pressure compounding, combined velocity and pressure compounding, velocity and pressure variation along the flow.
Reaction Turbine: Mechanical details, principle of operation, Thermodynamic analysis of a stage, Degree of reaction, velocity diagram, parson's reaction turbine, condition for maximum efficiency.

UNIT-IV:
Rotary Compressors: Working Principles of - Root's blower, vane blower and Screw compressor.
Centrifugal Compressors: Mechanical details and principle of operation, velocity and pressure variation, Energy transfer. Impeller blade shape-losses, slip factor, power input factor, pressure co-efficient and adiabatic co-efficient, velocity diagrams.
Axial Flow Compressors: Mechanical details and principle of operation, velocity triangles and energy transfer per stage, degree of reaction, work done factor, isentropic efficiency, pressure rise calculations, Polytrophic efficiency.

UNIT-V:
Gas Turbines: Classification of Gas Turbines, Ideal cycle, essential components, parameters of performance, actual cycle, regeneration, inter cooling and reheating, closed and semi closed cycles, merits and demerits, Combustion chambers and turbines for Gas Turbine plants.
Jet Propulsion: Principle of operation, Classification of Jet propulsion engines, working principles with schematic diagram and representation on T-s diagram, Thrust, Thrust power and propulsion efficiency. Needs and demands met by Turbo Jet Engines, Schematic diagram, Thermodynamic cycle, performance evaluation thrust augmentation methods.
Rockets: Application - working principle, Classification, Propellant type, Thrust, Propulsive efficiency – Specific impulse, solid and liquid propellant Rocket Engines.

TEXT BOOKS:

1. Thermal Engineering, Mahesh M. Rathore, 15th Edition, Tata McGraw-Hill, 2016
2. Thermal Engineering, R.K Rajput, 10th Edition, Lakshmi Publications, 2017
3. Gas Turbines, V. Ganesan, 3rd Edition, Tata McGraw-Hill, 2010

REFERENCES:

1. Fundamentals of Engineering Thermodynamics, Rathakrishnan, PHI, 2005
2. Thermal Engineering, P. L. Ballaney, 5th Edition, Khanna, 2010
3. Fundamentals of Turbo Machinery, B. K. Venkanna, Prentice Hall International, 2019
4. Thermodynamics and Heat Engines, R. Yadav, 7th Edition, Central Book Depot, 2007
5. Thermal Engineering, M. L. Marthur & Mehta, 3rd Edition, Jain Bros. Publishers, 2014

ONLINE RESOURCES:
1. https://archive.nptel.ac.in/courses/112/107/112107216/
2. http://ndl.iitkgp.ac.in/he_document/nptel/nptel/courses_112_107_112107216_video_lec26

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1ME401) COMPUTATIONAL FLUID DYNAMICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Fluid Mechanics, Heat Transfer

COURSE OBJECTIVES:
- To familiarize with the differential equations that from basis for CFD
- To understand different numerical methods involved in problem solving
- To formulate different kinds of physical problems with the different schemes and boundary conditions
- To apply the CFD techniques to physical problems

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Develop the fluid flow and heat transfer governing equations
CO-2: Apply and analyze different mathematical models and computational methods for flow and heat transfer simulations
CO-3: Conduct stability analysis and check the applicability of different schemes
CO-4: Solve computational problems related to fluid flow and heat transfer with FDM and FVM

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	3	3	3	-	3	-	-	-	-	2	1	2	2	2
CO-2	3	3	2	-	3	-	-	-	-	2	1	2	2	2
CO-3	3	3	3	-	2	-	-	-	-	2	1	2	2	2
CO-4	3	3	3	-	2	-	-	-	-	2	1	2	2	2

UNIT-I:
Introduction to Computational Fluid Dynamics: Computational Fluid Dynamics: Computational Fluid Dynamics as a Research Tool, Computational Fluid Dynamics as a Design Tool, The Impact of Computational Fluid Dynamics - some other examples: Automobile and Engine Applications, Industrial Manufacturing Applications, Civil Engineering Applications, Environmental Engineering Applications, Naval Architecture Application.

UNIT-II:
The Governing Equations of Fluid Dynamics: Introduction, Models of the Flow: Finite Control Volume, Infinitesimal Fluid Element, The Substantial Derivative (Time Rate of Change Following a Moving Fluid Element), The Divergence of the Velocity: Its Physical Meaning. The Continuity Equation: All Four models of flow, Integral versus Differential form of the Equations, The Momentum Equation, The Energy Equation, Equations for viscous flow (the Navier – stokes equation), Equations for Inviscid Flow (the Euler

Equations), Physical Boundary Conditions. Mathematical Behavior of Partial Differential Equations, Classification of PDEs.

UNIT-III:

Finite Difference Method: Taylor's series – Derivation of Finite Difference Formulae for Partial Derivative Terms - FD formulation of 1D Elliptic PDEs - 1D steady state heat transfer problems – Cartesian co-ordinate systems – boundary conditions. Explicit and Implicit Approaches: errors and analysis of stability

UNIT-IV:

Finite Volume Method: Finite Volume Method: Formation of Basic rules for Finite Volume approach – General Nodal Equation -Interface Thermal Conductivity — Treatment of Source Term and Treatment of Nonlinearity. Solution of 1D Elliptic PDEs - Heat conduction problems - Solution of 1D Parabolic PDEs – Explicit Method and Implicit Methods

UNIT-V:

FVM to Convection and Diffusion: General Form of Governing Equations for Fluid Flow and Heat transfer – Burger's equation - Steady 1D Convection Diffusion – Discretization Schemes and their assessment – Treatment of Boundary Conditions - Pressure Velocity Coupling - SIMPLE & SIMPLER (revised algorithm) Algorithms.

TEXT BOOKS:

1. Computational Fluid Dynamics: Basics with Applications, John D. Anderson, McGraw-Hill, 2015
2. Numerical Heat Transfer and Fluid Flow, Suhas V. Patankar, Hemashava Publishers Corporation & McGraw-Hill, 2018

REFERENCES:

1. Finite difference methods in heat transfer, Özişik, M. Necati, Helcio RB Orlande, Marcelo J. Colaço, and Renato M. Cotta. CRC press, 2017
2. Heat Transfer: A Basic Approach, Ozisik, M. Necati, McGraw-Hill, 1985
3. Fundamentals of Computational Fluid Dynamics, Tapan K. Sengupta, Universities Press, 2004
4. An Introduction to Computational Fluid Dynamics, H. K. Versteeg and W. Malalasekera, Pearson Education Limited, 2007
5. Computational Fluid Flow and Heat Transfer, Muralidaran, Narosa Publication, 2014

ONLINE RESOURCES:

1. https://nptel.ac.in/courses/112105045
2. https://ocw.mit.edu/courses/2-29-numerical-fluid-mechanics-spring-2015/

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1ME402) PRODUCTION PLANNING AND CONTROL

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Industrial Engineering and Management

COURSE OBJECTIVES:
- To understand the basic concepts of production planning
- To identify the different types of forecasting techniques
- To learn about the various scheduling methods
- To explore the process of routing and duties of dispatcher

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Define the objectives, functions, and elements of production planning and control
CO-2: Analyze the importance of forecasting in production systems
CO-3: Evaluate scheduling techniques
CO-4: Understand the procedures of routing, line balancing, dispatching and follow-up functions

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	3	2	1	-	-	-	-	-	-	-	-	-	3	2
CO-2	2	3	2	2	1	-	-	-	-	-	-	-	3	3
CO-3	3	3	2	2	2	1	-	-	-	-	-	-	3	3
CO-4	3	2	2	3	2	1	-	-	-	-	-	1	3	3

UNIT-I:
Introduction: Definition – objectives and functions of production planning and control – elements of production control – types of production – organization of production planning and control department – internal organization of department.

UNIT-II:
Forecasting: Importance of forecasting – types of forecasting, their uses – general principles of forecasting – forecasting techniques – qualitative methods and quantitative methods.

UNIT-III:
Routing: Definition – routing procedure –route sheets – bill of material – factors affecting routing procedure.
Scheduling: Definition, techniques, types, standard scheduling methods, Scheduling Techniques- Gantt Charts, LOB, Johnson's job sequencing rules- n jobs on 2machines, n jobs on 3 machines.

UNIT-IV:
Line Balancing: Introduction, objectives, terms related to line balancing, procedures, simple problems Aggregate planning-Introduction, Inputs to aggregate planning, strategies of aggregate planning, chase planning, expediting, controlling aspects.

UNIT-V:
Dispatching: Activities of dispatcher – dispatching procedure – follow up – definition – reason for existence of functions – types of follow up, applications of computer in production planning and control.

TEXT BOOKS:

1. Elements of Production Planning and Control, Samuel Eilon, Universal Book Corp., 2015
2. Manufacturing, Planning and Control, Partik Jonsson Stig-Arne Mattsson, Tata McGraw-Hill, 2009

REFERENCES:

1. Production Planning and Control, Mukhopadhyay, PHI, 2015
2. Production and Operations Management, Shailendra Kale, McGraw-Hill, 2017
3. Production and Operations Management, Ajay K. Garg, McGraw-Hill, 2019

ONLINE RESOURCES:
1. https://nptel.ac.in/courses/112107143
2. https://archive.nptel.ac.in/courses/110/107/110107141/

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1ME403) MACHINE LEARNING AND ITS APPLICATIONS IN MECHANICAL ENGINEERING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Matrices and Calculus, Ordinary Differential Equations, Partial Differential Equations, Probability and Statistics

COURSE OBJECTIVES:
- To introduce the concepts of Machine learning and need for data preparation
- To understand and apply linear and non-linear algorithms
- To understand and apply neural networks and ensemble algorithms

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Comprehend concepts of Machine Learning and need for data preparation
CO-2: Understand and apply linear algorithms
CO-3: Understand and apply Non-linear algorithms
CO-4: Analyse and apply neural networks, ensemble algorithms

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	3	3	3	3	2	1	2	2	2	2	1	2	3	3
CO-2	3	3	3	3	2	1	2	2	2	2	1	2	3	3
CO-3	3	3	3	3	2	1	2	2	2	2	1	2	3	3
CO-4	3	3	3	3	2	1	2	2	2	2	1	2	3	3

UNIT-I:
Introduction to Machine Learning and Data Preparation: Introduction to machine learning, Data Loading, scaling of data- normalize and standardize data, algorithm evaluation methods, evaluation metrics, simple examples on data from biomechanical and solid mechanics applications.

UNIT-II:
Linear Algorithms: Simple linear regression, multi variate linear regression, logistic regression, prediction using simple datasets on transient Heat transfer problems.

UNIT-III:
Non-Linear Algorithms: Classification and regression trees, Naïve Bayes, k-NN, predictions with simple datasets on manufacturing applications.

UNIT-IV:
Neural Networks: Introduction to Perceptron and Neural Networks, Activation functions, architecture, Types of Neural Networks- Single layer feed-forward network,

Multilayer feed-forward network, Multi-Layer Perceptron (MLP), Recurrent neural networks, Back propagation algorithm, Simple problems on Back Propagation.

UNIT-V:

Computer Vision: Introduction to Convolutional Neural Networks (CNNs), CNN's Basic architecture, simple image processing examples, applications in manufacturing.

Ensemble Algorithms: Introduction on ensemble methods, Bootstrap Aggregation, Random Forest, stacked generalization algorithms, simple examples on predicting mechanical properties on printed parts.

TEXT BOOK:

1. Machine Learning Algorithms From Scratch With Python, Brownlee Jason, Machine Learning Mastery, 2016

REFERENCES:

1. Artificial Intelligence: A Modern Approach, Stuart Russell & Peter Norvig, Prentice-Hall, 3rd Edition, 2009
2. Artificial Intelligence, Ela Kumar, Wiley, 2021
3. Artificial Intelligence: Concepts and Applications, Lavika Goel, Kindle Edition, Wiley, 2021
4. Nature-Inspired Optimization in Advanced Manufacturing Processes and Systems, Edited by Ganesh M. Kakandikar and Dinesh G. Thakur, 1st Edition, CRC Press, 2021

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1ME404) MECHANICS OF COMPOSITE MATERIALS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Mathematics, Physics, Chemistry, Engineering Mechanics & Mechanics of Solids

COURSE OBJECTIVES:
- To understand composite materials and their properties, relationship between them and manufacturing methods
- To understand the principles of material science applied to composite materials
- To study the equations to analyze problems by making good assumptions and learn systematic engineering methods to solve practical composite mechanics problems

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Apply fundamental knowledge of mathematics to model and analyze of composite materials
CO-2: Understand the manufacturing methods of the composite materials
CO-3: Recognize the behavior of composite materials
CO-4: Analyze the failure modes of composites

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	1	-	1	1	1	1	1	1	-	-	-	2	1	1
CO-2	1	1	1	1	2	1	2	2	-	-	-	2	1	1
CO-3	2	2	2	2	2	2	2	1	-	-	-	2	2	2
CO-4	2	2	2	2	3	2	2	1	-	-	-	2	2	2

UNIT-I:
Introduction: Definition – Classification and characteristics of Composite materials. Advantages and Challenges. Effect of reinforcement (size, shape, distribution, volume fraction, orientation) on overall composite performance. Nano composites, recent trends in composites.
Matrices: classification of matrices – polymers, metals and ceramics – properties and applications

UNIT-II:
Reinforcements: Preparation-layup, curing, properties and applications of glass fibers, carbon fibers, Kevlar fibers and Boron fibers. Properties and applications of whiskers, particle reinforcements. Carbon – Carbon composites.
Mechanical Behavior of Composites: Rule of mixtures, Inverse rule of mixtures. Isostrain and Isostress conditions.

UNIT-III:

Manufacturing of Metal Matrix Composites: Casting – Solid State diffusion technique, Cladding – Hot isostatic pressing. Properties and applications.

Manufacturing of Ceramic Matrix Composites: Liquid Metal Infiltration – Liquid phase sintering.

Manufacturing of Polymer Matrix Composites: Preparation of Moulding compounds and prepregs – hand layup method – Autoclave method – Filament winding method – Compression moulding – Reaction Injection moulding. Properties and applications, Introduction to Machining of Composites, Characterization of composites.

UNIT-IV:

Elastic Behavior of Laminate: Basic assumptions, Strain-displacement relations, Stress-strain relation of layer within a laminate, Force and moment resultant, General load–deformation relations.

UNIT-V:

Strength: Laminar Failure Criteria-strength ratio, maximum stress criteria, maximum strain criteria, interacting failure criteria, hygrothermal failure. Laminate first ply failure-insight, strength degradation; stress concentrations. Introduction to Sandwich Composites.

TEXT BOOKS:

1. Analysis and Performance of Fiber Composites, B. D. Agarwal, 3rd Edition, Wiley Publishers, 2017
2. Mechanics of Composite Materials, Robert M. Jones, 2nd Edition, Scripta Book Company

REFERENCES:

1. Material Science and Technology, Vol. 13-Composites, R. W. Cahn, VCH, West Germany, 1990
2. Materials Science and Engineering-An Introduction, W. D. Callister Jr., Adapted, R. Balasubramaniam, John Wiley & Sons, NY, Indian Edition, 2007
3. Composite Materials, K. K. Chawla, Springer, 2012
4. Composite Materials Science and Applications, Deborah D. L. Chung, 2nd Edition, Springer, 2010
5. Composite Materials Design and Applications, Danial Gay, Suong V. Hoa and Stephen W. Tsai, 4th Edition, CRC Press, 2022

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1ME405) MODERN MACHINING PROCESSES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Basics of Manufacturing Technology

COURSE OBJECTIVES:
- To Understand the conventional and non-conventional manufacturing terms
- To Learn about different types of non-conventional machining processes and advanced machines
- To select a suitable machining process for materials considering their merits and demerits

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Summarize the principle and processes of Mechanical machining
CO-2: Identify the principles, processes and applications of Electrochemical Machining & Thermo-Electrical Machining
CO-3: Apply modern welding processes for materials joining
CO-4: Relate the different types of micro-manufacturing processes

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	2	-	-	-	-	-	1	-	-	-	-	2	3	-
CO-2	2	-	-	-	3	2	1	-	-	-	-	2	3	-
CO-3	2	-	-	-	3	2	1	-	-	-	-	2	3	-
CO-4	1	-	-	-	2	2	1	-	-	-	-	2	3	-

UNIT-I:
Introduction: Limitation of conventional manufacturing processes, need for nonconventional manufacturing processes, and classification.
Mechanical Machining - Construction & principle of operation, advantages, limitations and applications of - Ultrasonic machining (USM), Abrasive jet machining (AJM), Water jet machining (WJM), Abrasive water jet machining (AWJM).

UNIT-II:
Electro Chemical Machining: Construction & principle of operation, advantages, limitations and applications of – Chemical Machining, Electrochemical machining (ECM) - Electrochemical drilling (ECD) - Electrochemical grinding (ECG) - Electrochemical honing (ECH) - Shaped Tube Electrolytic Machining (STEM).

UNIT-III:
Thermo-Electrical Machining: Construction & principle of operation, advantages, limitations and applications of – Electrical discharge machining (EDM) - Electrical

discharge wire cutting (EDWC) - Electron beam machining (EBM) – Ion Beam Machining (IBM), Plasma Arc Machining (PAM).

UNIT-IV:

Laser Machining: Laser types - Construction & principle of operation, advantages, limitations and applications of - Laser beam machining (LBM), Laser cutting (LC), Laser drilling (LD), Laser marking and engraving (LM), Laser micromachining (LMM).

UNIT-V:

Micro-Machining: Overview of Micro-manufacturing, Micro-products and design considerations for Manufacturing, Construction & principle of operation, advantages, limitations and applications of Micro-EDM, Micro-electrochemical machining, Micro-milling, Micro-USM.

TEXT BOOKS:

1. Advanced Machining Processes, V. K. Jain, Allied Publishers Ltd., 2009
2. Modern Machining Process, P. C. Pandey and H. S. Shan, McGraw-Hill, 2013
3. Manufacturing Engineering and Technology, Serope Kalpakjian and Steven R Schmid, 8th Edition, Prentice Hall, 2020

REFERENCES:

1. Nontraditional Manufacturing Processes, G. F. Benedict, CRC Press, T & F, 2019
2. Micro-Manufacturing Engineering and Technology, Yi Qin, Elsevier, 2010
3. Manufacturing Science, A. Ghosh and A. K. Mallik, Affiliated East-West Press Pvt., 2010
4. Laser Materials Processing, Steen W. M. and Watkins, K., 4th Edition, Springer, 2010
5. Nonconventional Machining, P. K. Mishra, Narosa Publishing House, 2007

ONLINE RESOURCES:

1. https://onlinecourses.nptel.ac.in/noc22_me119/preview

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC2ME411) PRODUCTION DRAWING PRACTICES LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Engineering Drawing/Graphics

COURSE OBJECTIVES:
- To learn the principles of machine drawing conventions
- To analyze the machine elements like screw threads, nuts, bolts, keys and riveted joints
- To understand the limits, fits and tolerances and represent them in drawings
- To discuss and understand the different views of part and assembly drawings and subsequently draw the views of assembly and part drawings respectively of machine & engine parts/components

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: apply the Knowledge of Materials and Machine Drawing Conventions to draw the various Machine Parts/Components/Elements
CO-2: analyse all the parts & assemble them and produce with views of the assembly
CO-3: use and represent limits fits and tolerances and represent heat and surface treatment symbols on drawings
CO-4: produce part and detailed drawings of various given assemblies using Auto-CAD Software

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	2	-	2	-	-	-	-	-	-	-	-	-	2	-
CO-2	-	2	-	2	2	-	-	2	-	-	-	-	-	-
CO-3	-	-	-	-	-	2	2	-	2	-	2	-	-	2
CO-4	-	-	-	-	-	-	-	-	-	2	-	2	-	-

UNIT-I:
Conventional Representations:
Need for drawing conventions – Introduction to IS conventions.
Conventional representation of important materials, common machine elements, welded joints, basic electrical, hydraulic and pneumatic circuits & symbols.

UNIT-II:
Drawing of Machine Elements and Simple Parts:
a) Popular forms of screw threads, bolts, nuts, stud bolts
b) Keys, cotter joints, knuckle joints, shaft couplings
c) Journal and collar bearings

UNIT-III:

Assembly Drawings:

Assembly drawings of Engine Eccentric, Screw Jack and Feed Check Valve.

UNIT-IV:

Limits, Fits, Tolerances, Surface Roughness and Heat Treatment Symbols:

a) Limits, Fits and Tolerances: Types of fits and estimation of limits using tables.

b) Form and Positional Tolerances: Introduction and indication of the tolerances of form and position on drawings.

c) Indication of Surface Roughness, Heat Treatment and Surface Treatment symbols used on drawings.

UNIT-V:

Part/Component Drawings:

Drawing of parts/components from the given assembly views of Stuffing Box, Petrol Engine Connecting Rod and Lathe Tail Stock with indication of sizes, tolerances, surface roughness, heat treatment, form and positional tolerances etc.

Note: Use Modelling Softwares like CATIA as a Tool.

TEXT BOOKS:

1. Machine Drawing, K. L. Narayana, P. Kannaiah and K. Venkata Reddy, New Age Publishers, 2016
2. Production and Drawing, K. L. Narayana & P. Kannaiah, New Age Publishers, 2015

REFERENCES:

1. Machine Drawing, Siddeswar, Kannaiah and Sastry, New Age Publishers, 2017
2. Geometric Dimensioning and Tolerancing, James D. Meadows, B. S. Publications, 2017

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC2ME412) COMPUTATION AND SIMULATION LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Basic Mechanical Engineering courses

COURSE OBJECTIVES:
- To model and simulate, mechanical systems using MATLAB and Simulink
- To analyze mechanical systems using MATLAB and Simulink in the kinematics, dynamics, control systems, vibrations, and thermodynamics,
- To use computational tools for real-world engineering applications

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: To model basic control system components. Simulation of first order and second-order systems
CO-2: To simulate multi-domain systems to integrate mechanical, electrical, and fluid components
CO-3: To design systems in Simulink
CO-4: To simulate of multi-domain systems

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	3	3	3	3	3	1	-	-	1	2	1	2	3	2
CO-2	3	3	3	3	3	1	-	-	1	2	1	2	3	2
CO-3	3	3	3	3	3	1	-	-	1	2	1	2	3	2
CO-4	3	3	3	3	3	1	-	-	1	2	1	2	3	2

LIST OF EXPERIMENTS:

Introduction to MATLAB & Simulink Basics: Modeling with Blocks
1. Simulate the dynamics of a simple pendulum/free bodies/rotating bodies and plot its position over time
2. Model and simulate the vibration of a spring-mass system and analyze its frequency response
3. Model and simulate a mass-spring-damper system using Simscape components
4. Build and simulate a rotating drum system and analyze its rotational dynamics
5. Model a heat exchanger and simulate its thermal performance under different conditions
6. Design and simulate a PID controller for controlling a mass position

7. Simulate a system with friction and compare the results with and without nonlinear elements
8. Model and simulate a 2D robotic arm with two links and analyze its kinematic and dynamic behavior
9. Optimize the design of a gear system using optimization algorithms
10. Simulate the stress and strain analysis of a mechanical structure using basic FEM principles in MATLAB
11. Design and simulate a robotic arm for a pick-and-place task using Simulink and Simscape Multibody.
12. Simulate a real-time system and implement the code on an embedded system and simulate its interaction with hardware

Software Tools:
- MATLAB
- Simulink
- Simscape

TEXT BOOKS:
1. MATLAB for Engineers, Holly Moore, 6th Edition, Pearson Education, 2022
2. MATLAB and Simulink for Engineers Agam Kumar Tyagi, Oxford University Press, 2012
3. Dynamic Systems and Control Engineering, Nader Jalili, Southern Methodist University; Nicholas W. Candelino, GE Research, Cambridge University Press, 2023

REFERENCES:
1. Modeling and Simulation of Mechatronic Systems using Simscape, Shuva Das, Morgan & Claypool Publishers, 2020
2. Mechanisms and Vibration Analysis with SolidWorks and MATLAB/Simscape, Dr. Cyrus Raoufi, CYRA Engineering Services, 2021

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1ME406) GAS DYNAMICS AND JET PROPULSION

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Fluid Dynamics and Thermodynamics

COURSE OBJECTIVES:

- To understand the basic governing equations of fluid flow
- To understand simplified studies of one-dimensional flow and certain special cases like Rayleigh flow and Fanno flow
- To understand the phenomenon of shock waves and its effect on flow.
- To understand concept of oblique shock as a consequence of two-dimensional flows along with various conditions
- To gain some basic knowledge about jet propulsion and Rocket Propulsion

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Apply the basic governing equations of fluid flow
CO-2: Analyse one dimensional flow and certain special cases like Rayleigh flow and Fanno flow
CO-3: Analyse shock waves and its effect on flow.
CO-4: Analyse oblique shock as a consequence of two dimensional flows along with various conditions
CO-5: To apply gas dynamics principles to jet and space propulsion systems

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	3	3	2	1	-	-	-	-	-	-	-	-	3	1
CO-2	3	3	2	1	-	-	-	-	-	-	-	-	3	1
CO-3	3	3	1	1	-	-	-	-	-	-	-	-	3	1
CO-4	3	3	1	1	-	-	-	-	-	-	-	-	3	1
CO-5	3	3	1	1	-	-	-	-	-	-	-	-	3	1

UNIT-I:
Introduction to Gas Dynamics: Control volume and system approaches acoustic waves and sonic velocity - Mach number - classification of fluid flow based on mach number - mach cone-compressibility factor - General features of one dimensional flow of a compressible fluid - continuity and momentum equations for a control volume.

UNIT-II:
Isentropic Flow of an Ideal Gas: Basic equation - stagnation enthalpy, temperature, pressure and density-stagnation, acoustic speed - critical speed of sound-dimensionless velocity-governing equations for isentropic flow of a perfect gas -

critical flow area - stream thrust and impulse function. Steady one dimensional isentropic flow with area change-effect of area change on flow parameters-chocking- convergent nozzle - performance of a nozzle under decreasing back pressure -De lavel nozzle - optimum area ratio effect of back pressure - nozzle discharge coefficients - nozzle efficiencies

UNIT-I:

Simple Frictional Flow: Adiabatic flow with friction in a constant area duct-governing equations - fanno line limiting conditions - effect of wall friction on flow properties in an Isothermal flow with friction in a constant area duct-governing equations - limiting conditions. Steady one dimensional flow with heat transfer in constant area ducts-governing equations - Rayleigh line entropy change caused by heat transfer - conditions of maximum enthalpy and entropy.

UNIT-IV:

Shocks and Quasi One Dimensional Analysis: Intersection of Fanno and Rayleigh lines. Shock waves in perfect gas- properties of flow across a normal shock - governing equations - Rankine Hugoniat equations - Prandtl's velocity relationship - converging diverging nozzle flow with shock thickness - shock strength, oblique shock waves

UNIT-V:

Air Craft and Space Propulsion: Theory of jet propulsion – Thrust equation – Thrust power and propulsive efficiency –Operation principle, cycle analysis and use of stagnation state performance of turbojet, turbofan and turbo prop engines.

Space Propulsion: Types of rocket engines – Propellants-feeding systems – Ignition and combustion – Theory of rocket propulsion – Performance study – Staging – Terminal and characteristic velocity – Applications – space flights.

TEXT BOOKS:

1. Modern Compressible Flow with historical perspective, John D. Anderson Jr., 4th Edition, McGraw-Hill International Edition, 2021
2. Fundamentals of Gas Dynamics, V. Babu, ANE Student Edition, 2008

REFERENCES:

1. Gas Dynamics and Jet Propulsions, Somasundaram P. R. S. L., New Age International Publishers, 1996
2. Aircraft and Missile Propulsion, Zucrow N. J., Vol. I & II, John Wiley, 1975
3. Principles of Jet Propulsion and Gas Turbines, Zucrow N. J., John Wiley, 1970
4. Dynamics and Thermodynamics of Compressible Fluid Flow, Shapiro A. H., John Wiley, 1953
5. Gas Turbines, Ganesan V., Tata McGraw-Hill, 1999

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1ME407) DESIGN FOR MANUFACTURING AND ASSEMBLY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Fundamentals of Manufacturing, Metallurgy & Material Science, Machine Tools and Metrology, Design Concepts, Automation.

COURSE OBJECTIVES:
- To impart the knowledge on steps involved in design process and material selection
- To understand about the design rules involved in machining and casting
- To understand about the design rules involved in metal joining, forging, extrusion and sheet metal work
- To understand about the design principles involved in manual and automatic assembly transfer systems

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Apply the knowledge on steps involved in design process and material selection
CO-2: Apply the knowledge on design rules involved in machining and casting
CO-3: Analyze the design rules involved in metal joining, forging, extrusion
CO-4: Design and analyze the principles involved in manual and automatic assembly transfer systems

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	2	2	2	1	-	3	3	1	3	-	-	-	3	1
CO-2	2	2	2	1	-	-	-	-	3	-	-	-	3	1
CO-3	2	2	2	1	-	-	-	-	3	-	-	-	2	1
CO-4	2	2	2	1	-	3	3	1	3	-	-	-	3	1

UNIT-I:
Introduction: Design philosophy, Steps in design process, General design rules for manufacturability, Basic principles of designing for economical production.
Creativity in Design: Design aspects covering environmental concerns and sustainability, power consumption, operational safety and fool proofing.
Materials: Selection of materials for design, commonly used metal sections, Criteria for material selection, Material selection interrelationship with process selection, Process Selection charts.

UNIT-II:
Machining Process: Overview of various machining processes, General design rules for machining, Dimensional tolerance and surface roughness, Design for machining

ease, Redesigning of components for machining ease with suitable examples, General design recommendations for machined parts.
Metal Casting: Appraisal of various casting processes, Selection of casting process, General design considerations for casting, Casting tolerances, Use of solidification simulation in casting design, Product design rules for sand casting.

UNIT-III:
Metal Joining: Appraisal of various welding processes, Factors in the design of weldments, General design guidelines - Pre and post treatment of welds, Effects of thermal stresses in weld joints, Design of brazed joints.
Forging and Extrusion: Design factors for Forging, Closed die forging design, Parting lines of die drop forging die design, General design recommendations. Design guidelines for extruded sections.

UNIT-IV:
Assembly Advantages: Development of the assembly process, Choice of assembly method, Assembly advantages, Social effects of automation, Overview of design for additive manufacturing.
Automatic Assembly Transfer Systems: Continuous transfer, Intermittent transfer, indexing mechanisms and an operator - paced free transfer machine.

UNIT-V:
Design of Manual Assembly: General design guidelines for manual assembly, Development of the systematic DFA methodology, Assembly efficiency, Classification system for manual handling, Classification system for manual insertion and fastening, Effect of part symmetry on handling time, Effect of part thickness and size on handling time, Effect of weight on handling time.

TEXT BOOKS:
1. Assembly Automation and Product Design, Geoffrey Boothroyd, 2nd Edition, CRC Pr I Llc, 2005
2. Engineering Design – Material and Processing Approach, George E. Dieter, 3rd Editon McGraw-Hill, 2000
3. Material Selection in Mechanical Design, Michael F. Ashby, 4th Edition Butterworth-Heinemann, 2010

REFERENCES:
1. Handbook of Product Design, Geoffrey Boothroyd, 2nd Edition, Marcel and Dekker,2002
2. Computer Aided Assembly Planning, A. Delchambre, Springer, 1992

ONLINE RESOURCES:
1. https://archive.nptel.ac.in/courses/107/103/107103012/
2. https://archive.nptel.ac.in/courses/112/101/112101005/

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1ME408) ADVANCES IN CAD/CAM

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Basic knowledge of CAD/CAM, Production Technology

COURSE OBJECTIVES:

- To comprehend the data exchange formats and know the different transformations in CAD modeling
- To understand the parametric representation of synthetic entities
- To compare the different representation schemes and comprehend the applications of CAD
- To understand the NC Systems, NC part programming fundamentals & CNC Systems
- To understand the concept of Adaptive Control, Computer Aided Inspection & Quality Control, and implementation of CAD/CAM software and Post Processor

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Asses the various data exchange formats used and object generation algorithms
CO-2: Derive and apply the parametric representation of synthetic curves and surfaces
CO-3: Creating various models using advanced modeling techniques.
CO-4: Work on NC & CNC systems and program
CO-5: Apply the concepts of AC and CAI & QC and implement CAD/CAM software and Post Processor

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	3	2	-	1	-	-	2	-	-	2	-	-	1	2
CO-2	2	2	2	2	1	-	-	-	-	-	1	2	2	-
CO-3	2	2	-	2	3	-	2	-	-	2	-	-	2	2
CO-4	2	2	1	2	-	2	-	-	-	-	2	-	1	2
CO-5	1	2	-	1	3	2	3	-	-	2	2	1	2	2

UNIT-I:
Basics of Computer Aided Design: Introduction to Computer Graphics, DDA and Bresenham's algorithm for generating various figures, Database structure for graphics modeling, Software Tools for CAD, Geometric Modelling, Mathematics of projections, Clipping, Properties calculation Hidden line and surface removal.
Graphics Standards: Graphics standards – IGES & STEP structure and implementation.

UNIT-II:
Parametric Representation of Synthetic Entities: Parametric representation: Hermite Cubic Spline, Bezier curve, B-Spline curve; Hermite Bi-cubic surface, Bezier surface, B-Spline surface, Knot Vector Generation, NURBS, COONs surface.

UNIT-III:
Advanced Modeling Applications: Feature Based and Parametric Modeling, Assembly Modeling – Bottom-Up and Top-Down Approach, Mechanical Assembly: Introduction, Assembly Modeling. Additive Manufacturing, Integration of CAD Tools with Industry 4.0 Technologies.

UNIT-IV:
NC Part Programming: Concept, format, codes, preparatory and miscellaneous codes, manual part programming, APT programming, macros, fixed cycles.
Post Processors for CNC: Introduction to post processors, necessity of a post-processor, the general structure of a post-processor, functions of a post-processor.

UNIT-V:
Adaptive Control: Adaptive control with optimization, Adaptive control with constraints, Adaptive control in machining processes – turning and grinding.
Computer Aided Inspection and Quality Control: CMM construction, Limitations of CMM, Computer Aided Testing, Optical inspection methods.

TEXT BOOKS:
1. CAD/CAM Theory and Practice, Ibrahim Zeid, McGraw-Hill International, 2009
2. Computer-Aided Design Manufacturing, K. Lalit Narayan, K. Mallikarjuna Rao and M. M. M. Sarcar, Prentice Hall of India, 2013
3. Computer Control of Manufacturing Systems, Yoram Koren, McGraw-Hill International, 1986

REFERENCES:
1. Mastering CAD/CAM, Ibrahim Zeid, McGraw-Hill International, 2005
2. CAD/CAM, P. N. Rao, Tata McGraw-Hill, 2017
3. Mathematical Elements for Computer Graphics, Roger D. F. and Adams A. McGraw-Hill, 1989
4. CAD/CAM Computer Aided Design and Manufacturing, Mikell P. Groover, E. W. Zimmers Jr., Prentice Hall of International, 2021
5. Computer Aided Manufacturing, T. C. Chang, Wysk, H. P. Wang, Pearson/ Prentice Hall International, 2020

ONLINE RESOURCES:
1. https://drmcet.digimat.in/nptel/courses/video/112102101/L29.
2. https://archive.nptel.ac.in/courses/112/102/112102101/

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1ME409) PRODUCT DESIGN AND PROCESS PLANNING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Engineering Design, Manufacturing Processes, CAD/CAM

COURSE OBJECTIVES:
- To introduce the basic concepts of product design and product development process
- To introduce the concepts of product architecture, industrial design and DFM in product development
- To understand concept of process planning
- To understand approaches and implementation techniques of process planning

COURSE OUTCOMES: After completion of the course, students should be able to
CO-1: Apply the product design and development process and the integration of product specifications and customer requirements in product development
CO-2: Implement concepts of product architecture, industrial design, DFM in product Development
CO-3: Apply the concept of computer aided process planning
CO-4: Implement Generative CAPP and Retrieval CAPP System

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	1	2	1	1	1	1	-	-	1	-	1	-	1	2
CO-2	2	2	2	2	1	1	-	-	1	-	1	2	1	1
CO-3	2	2	1	2	1	1	-	-	1	-	1	-	1	2
CO-4	2	2	1	2	1	2	-	-	1	-	2	-	1	2

UNIT-I:
Introduction to Product Development: Introduction to product design and development, Characteristics of successful product development, Composition of product development team, Challenges of product development, Generic product development process and its adaptation, Process flows for various product developments, Product development organizations, establishing the relative importance of customer needs.

UNIT-II:
Product Specifications: Definitions, when to establish specifications, establishing target specifications, setting final specifications
Concept Generation, Selection & Testing: Activity of concept generation, five step method, Introduction to concept selection, Benefits of structured method, Concept screening, Concept scoring, Concept testing methodology.

UNIT-III:

Product Architecture: Introduction, its implications, Establishing the architecture, Platform planning, Design issues

Industrial Design: Industrial design process – Need, Impact, Management and Assessment

Design for Manufacturing: Prototyping and Robust Design: DFM defined, Overview of DFM Process, Introduction to prototyping - Principles and Technologies, Planning for prototypes: Robust Design Process

UNIT-IV:

Introduction to CAPP: Introduction and definition of process planning, Scope of process planning, Information requirement for process planning system in CAD/CAM, Role of process planning, Advantages of conventional process planning over CAPP, structure of automated process planning system, Advantages and applications of CAPP, Computer programming languages for CAPP.

UNIT-V:

Approaches of Process Planning: Manual approach, CAPP approaches.

Generative CAPP System: Importance, Principle of Generative CAPP system, Knowledge based systems, Implementation, Benefits. Generative approach-Forward and backward planning,

Retrieval CAPP System: Significance, Group technology, Structure, Relative advantages, Implementation and applications. Examples of process planning system-CAM-I.

TEXT BOOKS:

1. Product Design and Development, Karl T. Ulrich and Steven D. Eppinger, 7th Edition, Tata McGraw-Hill, 2020
2. Automation, Production Systems, and Computer-Integrated Manufacturing, 5th Edition, Pearson, 2024
3. Computer Aided Design and Manufacturing, Dr. Sadhu Singh, Khanna Publishers,1998

REFERENCES:

1. Product Design, Kevin Otto and Kristin Wood, Pearson Education, 2000
2. Computer Aided Process Planning, H. P. Wang & J. K. Li, 1st Edition, Elsevier Science & Technology, 1991
3. Principles of Process Planning - A Logical Approach, Gideon Halevi and Roland D. Weill, Chapman & Hall, 1995

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1ME410) METAL CASTING TECHNOLOGY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Fundamentals of Manufacturing Processes

COURSE OBJECTIVES:
- To Learn various types of molding methods and casting techniques
- To impart the knowledge of Gates, gating system design and risers
- To understand the construction and working of various melting furnaces
- To know the fettling process and elimination of gases from castings
- To understand the various inspection methods of castings

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Select appropriate molding methods for a casting Process
CO-2: Design a casting - gating and riser system for a specific part
CO-3: Distinguish the various types of melting furnaces for different materials
CO-4: Assess the required fettling process for castings
CO-5: Apply various inspection methods to identify casting defects

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	2	2	1	-	-	1	-	1	2	2	1	3	1	1
CO-2	2	2	2	2	-	1	-	1	2	2	1	3	3	3
CO-3	2	2	2	-	-	1	1	1	2	2	1	3	2	2
CO-4	1	1	2	-	-	1	-	1	2	2	1	3	2	2
CO-5	2	1	1	-	-	1	-	1	2	2	1	3	3	2

UNIT-I:
Moulding Methods: Introduction to foundry and foundry sands, Moulding methods: Green sand moulding- dry sand moulding- no bake moulding- shell moulding-Investment casting- Permanent moulding- die casting and Centrifugal casting.
Modern Moulding Methods: Rheocasting- Thixocasting and Squeeze casting.

UNIT-II:
Gating and Risering: Elements of gating system, Types of gates and their functions Gating system and design: Pouring time and rate of flow, Risers: Riser functions, design considerations of riser, Solidification: solidification of pure metals and alloys

UNIT-III:
Melting: Selection and control of melting furnaces, Constructional details of Operation of crucible furnaces, Reverberatory furnaces- Rotary furnace – Core type and Coreless type Induction furnaces - Arc furnace (direct and indirect arc furnaces).

Fluidity: Measurement of fluidity; effects of various parameters on fluidity.

UNIT-IV:

Fettling: Removal of gates and risers, Grinding, Shot blasting and finishing, Casting defects, Remedies. Gases in Metal: Methods of elimination and control of dissolved gases in castings. Environment, Health and Safety aspects

UNIT-V:

Inspection and Quality Control: Review of X ray and gamma ray radiography; magnetic particle; Penetrant and ultrasonic inspections; Economics of casting

TEXT BOOKS:

1. Principles of Metal Casting, Heine R. W., Loper C. R. & Rosenthal P. C., Tata McGraw-Hill,1980
2. Principles of Foundry Technology, Jain P. L., Tata McGraw-Hill, 2017
3. Fundamentals of Metal Casting, Flinn, Addison Wesley,1963

REFERENCES:

1. Principles of Metal Casting, Heine, Loper & Rosenthal, McGraw-Hill,1980
2. Foundry Technology, Beeley P. R. Butterworth, 2002
3. Foundry Engineering, Srinivasan N. K., Khanna, 2001
4. Casting Vol. 15, ASM Metals Handbook, ASM International, 2008

ONLINE RESOURCES:

1. https://archive.nptel.ac.in/courses/112/107/112107083/

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1ME411) POWER PLANT ENGINEERING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Thermal Engineering and Basic Electrical Engineering

COURSE OBJECTIVES:
- To understand the layout of the different types of power plants
- To understand the concept of power from non-conventional source
- To apply the knowledge on various components in the power plants
- To understand the power plant economics and power distribution

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Analyze the working principal of the power plant, scope for future expansion
CO-2: Understand the concept on various equipment's used in the plant
CO-3: Evaluate the power plant economics and environmental consideration
CO-4: Apply the Knowledge to the power distribution and load factor importance

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	3	3	3	2	3	2	2	-	-	2	-	-	3	3
CO-2	3	2	2	-	-	2	2	-	2	2	2	2	3	3
CO-3	2	3	2	-	-	2	3	-	-	2	-	2	2	2
CO-4	3	3	3	2	2	1	1	-	-	2	-	2	3	2

UNIT-I:
Introduction to the Sources of Energy – Resources and Development of Power in India.
Steam Power Plant: Plant Layout, Working of different Circuits, Fuel and handling equipments, types of coals, coal handling equipments, and ash handling systems.
Combustion Process: Properties of coal – overfeed and underfeed fuel beds, pulverized fuel burning system and its components, combustion needs and draught system, cyclone furnace, Dust collectors, cooling towers and heat rejection.

UNIT-II:
Internal Combustion Engine Plant: Diesel Power Plant: Introduction – IC Engines, types, construction– Plant layout with auxiliaries – fuel supply system, air starting equipment, lubrication and cooling system – Cost of Diesel power Plant – Testing Diesel Power Plant Performance.

UNIT-III:
Gas Turbine Plant: Introduction – classification - construction – Layout with auxiliaries – Principles of working of closed and open cycle gas turbines. Combined Cycle Power Plants and comparison.

Hydro Electric Power Plant: Water power – Hydrological cycle / flow measurement – drainage area characteristics – Hydrographs – Types of hydroelectric power plants - storage and Pondage.
Hydro Projects and Plant: Classification – Typical layouts – plant auxiliaries – plant operation pumped storage plants - Estimation of power from a given catchment area – simple problems

UNIT-IV:
Power From Non-Conventional Sources: Utilization of Solar energy - Collectors- Fuel Cells Principle of Working, Wind Energy – types – HAWT, VAWT -Tidal Energy.
Nuclear Power Station: Nuclear fuel – breeding and fertile materials – Nuclear reactor –reactor operation.
Types of Reactors: Pressurized water reactor, Boiling water reactor, sodium-graphite reactor, fast Breeder Reactor, Homogeneous Reactor, Gas cooled Reactor, Radiation hazards and shielding – radioactive waste disposal.

UNIT-V:
Power Plant Economics and Environmental Considerations: Capital cost, investment of fixed charges, operating costs, general arrangement of power distribution, Load curves, load duration.
Curve: Definitions of connected load, Maximum demand, demand factor, average load, load factor, diversity factor – related exercises. Effluents from power plants and Impact on environment – pollutants and standardization for Environmental pollution standards – Methods of Pollution control. - Green House Effect – Green House Gases

TEXT BOOKS:

1. A Textbook of Power Plant Engineering, R. K. Rajput, 5th Edition, Laxmi Publications, 2016
2. Power Plant Engineering, G. R. Nagpal, 16th Edition Khanna Publishers, 1995

REFERENCES:

1. Power Plant Engineering: P. K. Nag, 4th Edition, McGraw-Hill Education, 2017
2. Power Plant Engineering, P. C. Sharma, 1st Edition, S. K. Kataria Publication, 2013
3. A Text book of Power Plant Engineering, S. Domkundwar and Arora, 8th Edition, Dhanpat Rai & Co., 2016
4. Power Station Engineering, E. I. Wakil, 1st Edition, McGraw-Hill, 2002
5. Irrigation and Water Power Engineering, B. C. Punmia, Ashok Kumar Jain, Arun Kumar Jain, 17th Edition, Lakshmi Publication, 2021

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1ME412) SUSTAINABLE DEVELOPMENT TECHNOLOGIES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Manufacturing, Operations Research

COURSE OBJECTIVES:
- To understand the principles of sustainable development and their relevance to mechanical engineering practices
- To acquaint with renewable energy technologies and sustainable manufacturing processes
- To gain knowledge of sustainability-related policies, frameworks, and strategies, particularly in the Indian context, for practical implementation
- To analyze and propose innovative solutions to address global and national sustainability challenges

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand the principles and importance of sustainable development in the context of mechanical engineering
CO-2: Evaluate and apply renewable energy and sustainable manufacturing practices
CO-3: Design systems with minimal environmental impact using sustainable technologies
CO-4: Analyze policies, frameworks, and practical solutions relevant to India's sustainability goals

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	2	2	2	1	1	2	3	2	2	1	2	2	3	2
CO-2	2	2	2	1	1	2	3	2	2	1	2	3	3	2
CO-3	2	1	3	1	2	2	3	2	2	1	2	3	2	3
CO-4	2	1	3	1	2	2	3	2	3	1	2	2	1	2

UNIT-I:
Fundamentals of Sustainable Development: Need & Scope of sustainable development, environmental, economic, and social dimensions of sustainability, applications of sustainable development in engineering, Overview of the UN Sustainable Development Goals (SDGs), sustainability-focused design, Industrial practices aligned with sustainability.

UNIT-II:
Renewable Energy Systems and Technologies: Solar thermal systems and photovoltaic technologies, Wind Energy and Wind turbine technologies,

Biomass conversion: Biogas, biofuels, and solid biofuels. Overview of tidal, geothermal, and small hydro projects, battery technologies and energy storage solutions, energy conservation in industries and transportation.

UNIT-III:
Sustainable Manufacturing and Design: Sustainable Manufacturing Principles, Waste reduction strategies: Reduce, reuse, recycle, Green Manufacturing Technologies, Role of additive manufacturing (3D printing). Sustainable materials and design for environment (DFE), Circular Economy, Closed-loop manufacturing. Strategies for implementing circularity in industries.

UNIT-IV:
Sustainable Transportation and Mobility: Green Vehicle Technologies, Electric vehicles (EVs), hybrid vehicles, Alternative Fuels Production and utilization of biodiesel, ethanol, and hydrogen Role of CNG and LNG, Smart Transportation Systems Integration of IoT in transport systems for efficiency.

UNIT-V:
Environmental Impact Assessment and Policies: Environmental Analysis Tools Life Cycle Assessment (LCA) of mechanical systems, Carbon footprint and water footprint assessment, ISO 14001 Environmental Management System. Key performance indicators for sustainable technologies, ESG report.

TEXT BOOKS:
1. Sustainable Engineering: Concepts, Design, and Case Studies, David T. Allen and David R. Shonnard, 1st Edition, Pearson, 2011
2. Renewable Energy Resources, John Twidell and Tony Weir, Routledge, 2021

REFERENCES:
1. Handbook of Sustainable Engineering, Joanne Kauffman, Kun Mo LEE, Springer, 2013
2. Renewables-based Technology: Sustainability Assessment, Dewulf Jo & Langenhove Herman Van (Eds) Chichester, John Wiley, 2006

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1ME413) ADVANCED MECHANICS OF SOLIDS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Mechanics of Solids, Engineering Mechanics, Material Science.

COURSE OBJECTIVES:
- To demonstrate concepts of three-dimensional stress strain analysis, failure theories under elastic and plastic deformation in structural components
- To learn and analyze concepts on unsymmetrical bending behavior of beams
- To apply concepts of stress analysis for designing curved beams and torsional members
- To explain method of evaluating deformations in beams for real-world elastic-plastic bending and yielding problems

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Analyze 3D stresses, transformations, and evaluate material failure criteria effectively
CO-2: Understand unsymmetrical bending, shear stresses, and analyze thin-walled beam behavior
CO-3: Analyze circumferential and radial stresses, deflections, and plastic behavior of curved and torsional beams
CO-4: Understand beam behavior on elastic foundations under various loading conditions

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	3	3	3	2	2	1	1	1	1	2	1	2	3	2
CO-2	3	3	3	2	2	1	1	1	1	2	1	2	3	3
CO-3	3	3	2	3	2	1	1	1	1	2	2	2	3	3
CO-4	3	3	3	3	3	1	1	1	2	2	2	2	3	3

UNIT-I:
Three Dimensional Stresses:
Introduction, Stress and Strains in 3-D – Cauchy's formula, Principal Stress,
Hydrostatic stress, deviatoric stress, stress transformations, Mohr circle, octahedral shear stress, strain energy densities, etc.
Theories of failure Overview of elastic theory of failurs- alternative yield criteria – mohr-coulomb yield criterion, Drucker-Prager yield criterion, and Hill's criterion for orthotropic materials; General yielding – elastic-plastic bending, fully plastic moment, shear effect on inelastic bending, modulus of rupture, comparison of failure criteria, and interpretation of failure criteria for general yielding.

UNIT-II:

Unsymmetrical Bending: Introduction; Doubly symmetric beams with skew loads; Pure bending of unsymmetric beams; Generalized theory of pure bending; Bending of beams by lateral loads; Shear centre; Shear stresses in beams of thin-walled open cross sections; Shear centers of thin walled open sections; General theory for shear stresses.

UNIT-III:

Bending of Curved Beams: Introduction; Circumferential stresses in a curved beam – location of neutral axis of cross section; Radial stresses in curved beams – curved beams made from anisotropic materials; Correction of circumferential stresses in curved beams having I, T, or similar cross sections – Bleich's correction factors; Deflections of curved beams – cross sections in the form of an I, T, etc.; Statically indeterminate curved beams – fully plastic versus maximum elastic loads for curved beams.

UNIT-IV:

Torsion: Torsion of Non-circular members, hollow members, thin walled sections; Membrane Analogy.

UNIT-V:

Beam on Elastic Foundations: General theory; Infinite beam subjected to concentrated load: boundary conditions – method of superposition, and beam supported on equally spaced discrete elastic supports; Infinite beam subjected to a distributed load segment – uniformly distributed load; semi infinite beam subjected to loads at its end; semi infinite beam with concentrated load near its end; Short beams.

TEXT BOOKS:

1. Advanced Mechanics of Materials, Arthur P. Boresi and Richard J. Schmidt, John Wiley, 2009
2. Mechanics of Materials, J. M. Gere and S. Timoshenko, 2nd Edition, CBS, 2004

REFERENCES:

1. Strength of Materials (Part 2): Advanced Theory and Problems, Stephen Timoshenko, 3rd Edition, CBS, 2015
2. Engineering Mechanics of Solids, E. P. Popov, Pearson Education, 2022
3. Strength of Materials Schaum's Series, Merle C. Potter, William Nash, 7th Edition, McGraw-Hill, 2019

ONLINE RESOURCES:

1. https://archive.nptel.ac.in/courses/112/101/112101095/
2. https://archive.nptel.ac.in/courses/112/102/112102284/

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1ME414) ELECTRIC AND AUTONOMOUS VEHICLES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Thermodynamics, IC Engine, Principles of Automobile Engineering

COURSE OBJECTIVES:

- To study the need of electric and hybrid vehicles for the society
- To understand about electric and hybrid vehicles, power transmission
- To provide various autonomous vehicles and batteries
- To understand the principle, working, characteristics of engine

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand the importance of electric and hybrid vehicles for society
CO-2: Justify the various types of electric and hybrid vehicles and power transmission
CO-3: Understand the various autonomous vehicles and batteries
CO-4: Discuss the power, Suspension, characteristics of engine

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	3	3	2	3	2	1	2	1	1	1	1	2	3	-
CO-2	3	3	2	2	2	1	2	1	1	1	1	2	3	-
CO-3	3	3	1	1	2	1	2	1	1	1	1	2	3	-
CO-4	3	3	2	1	2	1	2	1	1	1	1	2	3	-

UNIT-I:
Introduction: Sustainable Transportation - Population, Energy, and Transportation - Environment - Economic Growth – Emissions regulations and norms- impact of modern drivetrains on energy supplies-New Fuel Economy Requirement Emergence of Electric Vehicles- Basics of the EV - Constituents of an EV -Vehicle and Propulsion Loads.

UNIT-II:
Electric Vehicle: Introduction: Limitations of IC Engines as prime mover, History of EVs, EV system, components of EV-DC and AC electric machines: Introduction and basic structure-Electric vehicle drive train-advantages and limitations, Permanent magnet and switched reluctance motors-EV motor sizing: Initial acceleration, rated vehicle velocity, Maximum velocity and maximum gradability.

UNIT-III:
Hybrid Vehicle: Configurations of hybrids, advantages and limitations-Hybrid drive trains, sizing of components Initial acceleration, rated vehicle velocity, Maximum

velocity and maximum gradability Hydrogen: Production-Hydrogen storage systems-reformers

UNIT-IV:

Autonomous Vehicles: Brief Introduction of Autonomous Vehicles, Types of Autonomous Vehicles, architecture of Autonomous Vehicle, advantages and disadvantage of AV, Sensor Technology for AV, Roll of AI in Autonomous Vehicles.

Batteries: Types of batteries, constructional details, thermal management of batteries, vent management system, Battery life analysis, Battery performance degradation analysis.

UNIT-V:

Vehicle Systems: Transmission system: Need, Torque Speed Characteristics of IC Engine and Motor, Comparison of IC with EV: Transmission system, Selection of transmission system, Estimation of gear ratio, Differential, Brake system, Steering system, Suspension system.

TEXT BOOKS:

1. Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design, Mehrdad Ehsani, Yimin Gao, Sebastien E. Gay and Ali Emadi, CRC Press, 2004
2. Electric Vehicle Technology-Explained, James Larminie and John Loury, John Wiley, 2003

REFERENCES:

1. Electric and Hybrid Vehicles – Design Fundamentals, Iqbal Husain, CRC Press, 2010
2. Electric and Hybrid – Electric Vehicles, Ronald K. Jurgen, SAE, 2002
3. Light Weight Electric/Hybrid Vehicle Design, Ron Hodkinson and John Fenton, Butterworth–Heinemann, 2000
4. Hybrid Electric Vehicles Principles and Applications with Practical Perspectives, Chris Mi, M. Abul Masrur John Wiley, 2018,
5. Autonomous Vehicles Technologies, Regulations, and Societal Impacts, George Dimitrakopoulos, Aggelos Tsakanikas, Elias Panagiotopoulos, Elsevier Publications, 2021

ONLINE RESOURCES:

1. https://www.classcentral.com/course/swayam-engineering-graphics-5305
2. https://www.mooc-list.com/tags/engineering-drawing
3. https://nptel.ac.in/courses/108/103/108103009/

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1ME415) COMPUTER INTEGRATED MANUFACTURING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Computer Aided Design, Computer Aided Manufacturing, Machine Tools, Operations Research

COURSE OBJECTIVES:
- To understand planning required in manufacturing area now a days
- To learn the fundamentals of computer assisted numerical control programming
- To learn quality control and material handling
- To learn the guidelines and criteria for implementing CAD/CAM systems and assisted software's for manufacturing

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand the basics of manufacturing and application of group technology
CO-2: Develop CAPP systems and understand layouts of manufacturing systems
CO-3: Apply concept of quality control and material handling
CO-4: To introduce the students to concepts of Industry 4.0 leading to Smart Factory

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2
CO-1	1	-	-	-	2	1	-	-	-	-	-	-	1	1
CO-2	3	-	2	-	2	1	-	-	2	-	-	2	1	2
CO-3	2	-	2	-	3	1	-	-	2	-	-	-	2	2
CO-4	3	-	3	-	2	1	-	-	2	-	-	2	2	1

UNIT-I:
Introduction: Introduction to Product life cycle management, Need for CAD/CAM integration through computers, Benefits of integration.

UNIT-II:
Group Technology: Part Families, methods for developing part families. Parts Classification and Coding: Features of Parts Classification and Coding Systems, Opitz Parts Classification and Coding Systems, Role of group technology in CAD/CAM integration.

UNIT-III:
Computer Aided Process Planning (CAPP): Process planning, approaches to computer aided process planning, Types of CAPP, role of process planning in CAD/CAM integration.
Flexible Manufacturing Systems: Introduction to flexible manufacturing systems, Types of FMS, Types of FMS Layouts, advantages and disadvantages of FMS.

UNIT-IV:

Computer Aided Quality Control: Terminology in quality control, contact inspection methods, non-contact inspection methods, Computer Aided Testing, Integration of CAQC with CAD/CAM.

UNIT-V:

Computer Integrated Manufacturing Systems: CIM: Introduction of CIM, concept of CIM, evolution of CIM, CIM wheel, Benefits, integrated CAD/CAM.

TEXT BOOKS:

1. Automation, Production Systems and Computer Integrated Manufacturing, Mikell P. Groover, Pearson Education, 2014
2. Computer Aided Process Planning, H. P. Wang & J. K. Li, 1st Edition, Elsevier, 1991
3. Systems Approach to Computer Integrated Manufacturing, Nanua Singh, Wiley, 1996

REFERENCES:

1. Flexible Manufacturing Systems, Shivanad H. K., Benal M. M., Koti V., New Age International, 2006
2. Computer Aided Design and Manufacturing, David D. Bedworth, Mark R. Henderson, Philip M. Wolfe, McGraw-Hill, 1991
3. CAD/CAM, Mikel P. Groover, Emery W. Zimmer, Pearson Education, 2013
4. Handbook of Flexible Manufacturing Systems, Jha N. K., Academic Press, 2012

ONLINE RESOURCES:

1. https://www.classcentral.com/course/youtube-computer-integrated-manufacturing-47315
2. https://www.classcentral.com/course/swayam-computer-integrated